

Feisi Fu

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SUMMARY

Machine Learning Engineer with a **Ph.D. in Systems Engineering**, specializing in **large-scale retrieval and ranking**, **retrieval-augmented generation (RAG)**, and **LLM-based semantic search**. Currently leading production retrieval systems for **TikTok E-commerce Search** under strict latency constraints, delivering measurable gains in **recall quality**, **query understanding**, and **orders**.

EXPERIENCE

- **Machine Learning Engineer, TikTok (E-commerce Search Team)** *Apr. 2024 – Present*
 1. Led the evolution of **core embedding-based retrieval models** serving **thousands of QPS** by identifying and resolving training-serving mismatches across **mixed traffic scenarios**; redesigned optimization objectives, **false-negative masking**, and model architecture, improving **retrieval-exclusive orders by 294%** and **online orders by 3.4%**.
 2. **Designed and productionized GIST Token + Doc KV Cache** serving for a **Qwen2.5-0.5B relevance model**, compressing Doc inputs from **100 to 20 tokens**. The combined **GIST, Qwen2.5, and GQA optimizations** reduced per-Doc KV cache from **5 MB to 0.5 MB** and read latency from **30+ ms to 8 ms** without significant performance regression, while doubling per-GPU throughput and cutting GPU capacity needs by about **50%**.
 3. Built and productionized a **LLM-based semantic retrieval system** for **long-tail query understanding** by combining product text, **category attributes**, and **sequential user behavior**, improving understanding of broad **e-commerce intent queries** under **strict latency constraints**; **reduced query reformulation rate by 2.1%** and increased **online orders by 1.3%**.
- **Machine Learning Engineer Intern, TikTok (Infrastructure Team)** *Sep. 2023 – Nov. 2023*

Extended **LLMs training framework** to support PyTorch **Fully Sharded Data Parallel (FSDP)** and distributed checkpoint.
- **Research Scientist Intern, Meta (PyTorch ArchOpt Team)** *Aug. 2022 – Dec. 2022*

Integrated a new **quantization-aware training (QAT)** feature into large-scale PyTorch training pipelines, enabling activation quantization to **reduce training memory usage**.

SKILLS

- **Modeling:** Retrieval & Ranking, Learning-to-Rank, Contrastive Learning, Multi-Task Learning, Hard Negative Mining, LLM-based Retrieval
- **Systems:** Distributed Training, Low-Latency Serving, Training-Serving Consistency, Offline/Online Evaluation, Monitoring & Alerting, A/B Testing, Agentic Workflows
- **Programming:** Python, C++, PyTorch, TensorFlow

EDUCATION

- **Boston University** *Ph.D. in Systems Engineering, 2024*
- **Chinese Academy of Sciences** *M.S. in Mathematics, 2018*

SELECTED PUBLICATIONS

- **Sound and Complete Neural Network Repair with Minimality and Locality Guarantees.** *ICLR 2022.*
- **REGLO: Provable Neural Network Repair for Global Robustness Properties.** *AAAI 2024.*